

NOVA: QUERY-OCCURRENCE BINDING VIA LOCALLY VALID TIMESTAMP CONTRASTS FOR AUDIO TEMPORAL GROUNDING

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ABSTRACT

Audio temporal grounding must identify which occurrence answers a query when several acoustically valid windows coexist. Event presence alone does not determine which occurrence matches the requested relation. We introduce NOVA, combining relation-aware candidate generation, audio verification, and selective boundary refinement. Its core training module, RelTwin, learns query–occurrence binding through locally valid timestamp contrasts: shared excerpts form opposite-order windows whose correct and incorrect roles reverse with the query. Competing timestamp answers therefore differ in the requested relation, not acoustic validity. RelTwin reaches 59.43 mIoU on SpotSound-Bench and improves over matched SFT by 1.02 points, while joint localization improves by 31.25 points on synthetic relation-development data. NOVA reaches 86.86 mIoU on Clotho-Moment and 73.46 on the released UnAV subset. The binding advantage persists under timing intervention, supporting explicit supervision of which occurrence answers the query.

Index Terms— Audio temporal grounding, timestamp contrasts, query–occurrence binding, paired supervision

1. INTRODUCTION

Audio temporal grounding maps a query to event intervals [1, 2]. In a recording containing both “dog then rooster” and “rooster then dog,” the descriptions refer to different windows. Recognizing the categories establishes presence; answering the query requires selecting the requested order. Returning the same real window for both queries is a *query–occurrence binding error*: the wrong local occurrence is selected.

We propose NOVA, combining candidate learning for occurrence selection, audio verification for acoustic support, and refinement for boundary alignment. Our core training contribution is RelTwin (Fig. 1), which introduces *locally valid timestamp contrasts*. Shared excerpts form opposite-order windows with real acoustic support; the query determines which window answers it. Sequence supervision

rewards correct answers individually, while candidate supervision compares both answers under each query. The trained generator directly outputs timestamps.

We formulate query–occurrence binding as a choice between locally valid timestamp answers. NOVA is evaluated on SpotSound, Clotho-Moment, and UnAV; matched comparisons and inverse-query tests show improved occurrence selection, which persists under timing intervention.

2. RELATED WORK

Audio temporal grounding. Weakly supervised grounding learns temporal correspondences from clip-level captions [3]. Pengi [4], Qwen2-Audio [5], and Audio Flamingo 3 [6] provide audio-conditioned generation. TimeAudio adds temporal markers and absolute time-aware encoding [7]. SpotSound interleaves timestamps with audio tokens and uses absent-event queries to suppress hallucinated intervals [8], combining presence supervision with timestamp prediction. RelTwin targets the query-specific choice between occurrences when inverse relations both have acoustic support.

Compositional audio–language learning. CLAP aligns audio and text representations [9]; T-CLAP adds order-reversed descriptions and temporal-focused contrast [10]. CompA evaluates event order and attribute binding, while CompA-CLAP trains with composition-aware negatives and modular contrast [11]. CoSTALA combines spatio-temporal contrast, local event–caption alignment, and feature consistency [12]. These methods organize compositional audio–text representations. RelTwin instead compares the query-conditioned likelihoods of complete timestamp answers, supervising selection in a grounding model’s output space.

Hard negatives for temporal grounding. For compositional video grounding, SHINE combines generated hard-negative queries with within-video and cross-query saliency ranking [13], comparing segment relevance and query compositions. RelTwin’s candidates are complete timestamp answers within one recording. Each denotes a real event sequence; reversing the query exchanges their labels while preserving the audio. Sequence likelihoods train this query-specific choice.

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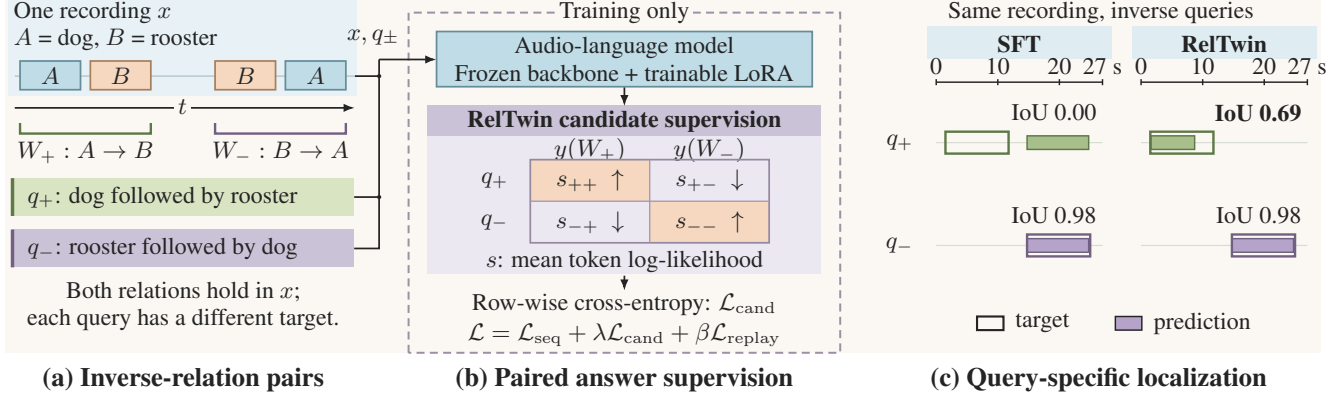


Fig. 1. RelTwin binds inverse queries to different windows in one recording. (a) Reused excerpts form two local targets. (b) The dashed enclosure marks training-only computation; shaded diagonal cells identify correct timestamp answers, and up/down arrows indicate training preferences. (c) In a relation-development example, SFT repeats one answer while RelTwin selects the corresponding window for each query. Outlines mark targets; filled bars show predictions, evaluated by set-IoU.

3. RELTWIN: LOCALLY VALID TIMESTAMP CONTRASTS

3.1. Two true relations, two different answers

In one recording x , RelTwin arranges the same ESC-50 excerpts [14], A and B , as $A \rightarrow B$ and $B \rightarrow A$. Their local targets W_+ and W_- run from the first excerpt’s onset to the second’s offset, including the gap. Queries q_+ and q_- request the respective orders. Both windows have acoustic support, but W_- is positive for q_- and negative for q_+ ; reversing the query exchanges their labels.

The templates “followed by” and “and then” yield two inverse-query pairs per recording. Earlier-order layouts alternate (Section 5.3). Category rehearsal targets the union of both occurrences of A or B , retaining category-grounding supervision.

3.2. Contrast timestamp answers during training

We serialize an interval set S as comma-separated spans, each formatted like from 1.500s to 11.750s (three-decimal precision). Its supervised assistant-completion tokens $y(S)$ exclude the prompt and are scored by

$$s_\theta(x, q, S) = \frac{1}{|y(S)|} \sum_t \log P_\theta(y_t | x, q, y_{<t}). \quad (1)$$

The 2×2 matrix M , with entries $M_{ij} = s_\theta(x, q_i, W_j)$ for $i, j \in \{+, -\}$, indexes queries by rows and answers by columns. Its diagonal labels $Y = I_2$ identify the correct query-answer pairs (Fig. 1b). Sequence supervision minimizes $\mathcal{L}_{\text{seq}} = -(M_{++} + M_{--})/2$, rewarding the two correct responses individually. It does not explicitly compare them with the other locally valid window. RelTwin trains the relative preferences $M_{++} > M_{+-}$ and $M_{--} > M_{-+}$ through

row-wise probabilities $p_i(j) = \text{softmax}_j(M_{ij}/\tau)$, where $\tau > 0$. The candidate loss is

$$\mathcal{L}_{\text{cand}} = -\frac{1}{2}[\log p_+(+) + \log p_-(-)]. \quad (2)$$

For a separately sampled category example (x_r, q_r, S_r) , the rehearsal loss is $\mathcal{L}_{\text{replay}} = -s_\theta(x_r, q_r, S_r)$. Together,

$$\mathcal{L} = \mathcal{L}_{\text{seq}} + \lambda \mathcal{L}_{\text{cand}} + \beta \mathcal{L}_{\text{replay}}. \quad (3)$$

Paired SFT sets $\lambda = 0$; RelTwin uses $\lambda > 0$ with identical relation and rehearsal data. Both directly generate timestamps at inference, without candidates or an additional scorer.

3.3. Joint localization and relation selection

For interval unions, $\text{IoU}_{\text{set}}(S, G) = |S \cap G|/|S \cup G|$ (zero for an empty union). Given predictions $\widehat{W}_i$, define $u_{ij} = \text{IoU}_{\text{set}}(\widehat{W}_i, W_j)$. PairAcc requires $\min(u_{++}, u_{--}) \geq 0.5$. Swap error occurs if $u_{++} \leq u_{+-}$ or $u_{--} \leq u_{-+}$, including ties. JointPairAcc requires PairAcc success without Swap error. We report pair-level percentages; same-answer counts use exact equality of parsed interval lists.

Predicting the union of two disjoint, equal-length targets for both queries passes PairAcc (IoU 0.5) but fails Joint-PairAcc. Inspired by paired compositional evaluation [15], JointPairAcc distinguishes query-specific selection from coverage.

4. EXPERIMENTAL SETUP

Data and evaluation protocol. Adaptation uses 40 ESC-50 classes: 256 recordings, 1,024 relation queries (512 pairs), and 512 rehearsal queries. Ten held-out classes provide 80 recordings, 320 queries, and 160 relation-development pairs, with disjoint original Freesound IDs. System evaluation uses

Table 1. Audio temporal grounding (%; higher is better). Baselines are from SpotSound Table 3 [8]; NOVA settings are in Section 4.1. Bold marks column maxima; deltas use per-metric prior main-table bests. †NOVA uses the released 100-query UnAV subset; reference scores follow the source paper’s evaluation protocol (Section 4).

Method	SpotSound-Bench			Clotho-Moment			UnAV subset [†]		
	mIoU	R1@.3	R1@.5	mIoU	R1@.3	R1@.5	mIoU	R1@.3	R1@.5
<i>Task-specific grounding models</i>									
WTATG	38.4	53.0	34.5	9.1	12.1	6.3	38.4	53.0	37.0
AM-DETR	19.5	25.5	15.0	80.9	89.8	88.0	42.8	59.0	46.0
<i>Proprietary audio–language models</i>									
Gemini-2.5-Flash	25.7	34.8	27.8	36.9	45.7	33.7	35.6	48.0	37.0
Gemini-2.5-Pro	19.7	20.0	17.3	32.5	40.4	32.5	34.6	39.0	35.0
<i>General-purpose open audio–language models</i>									
Kimi-Audio	3.2	3.0	1.3	0.9	0.7	0.1	5.3	8.0	2.0
Qwen2-Audio	4.4	5.5	1.8	5.7	6.1	1.3	9.7	14.0	4.0
Audio Flamingo 3	7.6	8.8	2.8	22.6	32.9	21.8	25.0	35.0	24.0
<i>Temporal grounding-enhanced models</i>									
TimeAudio	10.2	7.8	1.8	28.6	39.5	24.9	16.0	21.0	7.0
SpotSound-Q	49.9	65.0	51.3	85.4	93.6	91.2	72.4	88.0	74.0
SpotSound-A	57.9	76.5	59.5	85.6	93.4	91.0	69.8	86.0	74.0
NOVA	59.39	79.25	62.50	86.86	93.49	90.84	73.46	89.00	80.00
Prior main-table best	57.9	76.5	59.5	85.6	93.6	91.2	72.4	88.0	74.0
Δ vs. prior best	+1.49	+2.75	+3.00	+1.26	−0.11	−0.36	+1.06	+1.00	+6.00

400 SpotSound queries, 6,649 Clotho-Moment queries, and the released 100-query/77-audio UnAV subset [2]. UnAV scores clip intervals to audio duration (unclipped mIoU: 73.39). SpotSound reports 997 UnAV queries; our evaluation uses the released 100-query subset. Development and public-benchmark scores were used during method development.

Training settings. SFT and RelTwin initialize from SpotSound-A on Audio Flamingo 3. LoRA [16] uses rank 8, $\alpha = 16$, dropout 0.1, and 20.19M parameters. Both run 256 AdamW [17] updates at learning rate 5×10^{-6} , zero weight decay, gradient clipping 1.0, $\tau = 1$, $\beta = 0.5$, and RelTwin $\lambda = 1$. Each update accumulates one inverse-query pair and one rehearsal query. The matched comparison shares initialization, data order, code, environment, and optimizer updates. RelTwin additionally computes four gradient-free candidate scores followed by weighted gradient recomputation, versus SFT’s two positive-answer backpropagations. Both include rehearsal.

Inference. SFT and RelTwin are evaluated after 256 updates. All generators use 16 kHz mono audio, bfloat16, and greedy one-beam decoding without sampling, capped at 128 new tokens. The post-development timing diagnostic uses the frozen matched checkpoints (Section 5.3).

Reference models. Table 2 includes paired SFT and the official backbone. The backbone scores 58.42 mIoU versus published 57.9; RelTwin’s three-seed difference against it is +0.75 points (audio-group CI $[-0.24, 1.74]$).

Result reporting and statistics. RelTwin point results use seed 0, selected by SpotSound mIoU from five runs, with

its matched SFT control. Table 2(c) separately summarizes seeds 0/1/2 as mean $\pm$ sample SD. Training comparisons use 20,000 audio-group bootstrap resamples for 95% CIs. Relation comparisons resample 39 connected source-sharing groups (largest: 18 audios). CIs quantify audio-sample uncertainty conditional on the evaluated checkpoints; model selection is fixed across resamples.

4.1. NOVA system configurations

NOVA (Necessity-Oriented Verification over Audio) combines candidate generation, verification, and refinement. SpotSound uses RelTwin data with Jensen–Shannon consistency between p_+ and label-swapped p_- added to Eq. 3. Setwise preference optimization (SetPO) continues for 64 updates at 2×10^{-6} over six candidates: both windows, their expansions, enclosing span, and union. Soft targets combine set-IoU and interval-coverage quality. The RelTwin training comparison in Table 2 uses the three-term objective in Eq. 3.

Official, SFT, and SetPO seed-1 SpotSound predictions enter a LongNeedle-ESC50-fitted ridge router, using geometry and frozen query-existence scores after keeping/removing candidate intervals. Clotho and UnAV apply refinement to official predictions, whose initial mIoUs are 86.8543 and 73.4513. Refinement uses frozen SetPO/SpanTool features and ridge edit utilities: dynamic programming selects compatible boundaries, accepted when the 200-model bootstrap 5th percentile exceeds 0.02. Event count is preserved; radius $\min(0.25\text{ s}, 0.625\text{ event duration})$ is shared. AudioGrounding/AEGBench development evaluations selected the dura-

Table 2. Query–occurrence binding on the synthetic relation-development set and transfer to SpotSound. Panel (b) matches training data and updates; panel (c) reports replication means $\pm$ sample SD (Section 4). Joint denotes JointPairAcc; same-answer counts identical timestamp answers to inverse queries. Scores are percentages except pair counts.

Method	SpotSound	Relation development (160 inverse-query pairs)			
	mIoU $\uparrow$	mIoU $\uparrow$	Joint $\uparrow$	Swap error $\downarrow$	Same-answer pairs $\downarrow$
<i>(a) Backbone reference: shared evaluation protocol</i>					
SpotSound-A	58.42	37.64	0.63	65.00	15/160
<i>(b) Matched training comparison</i>					
Paired SFT	58.42	74.58	53.13	40.00	37/160
RelTwin	59.43	88.55	84.38	7.50	1/160
RelTwin – Paired SFT	+1.02	+13.97	+31.25	–32.50	–36 pairs
<i>(c) RelTwin replication: three training seeds</i>					
RelTwin	59.16 $\pm$ 0.26	87.05 $\pm$ 1.43	81.46 $\pm$ 2.53	8.13 $\pm$ 0.63	(1, 0, 0)/160

tion coefficient before these applications. Table 1 reports these NOVA configurations.

5. RESULTS AND ANALYSIS

5.1. Grounding performance across benchmarks

NOVA reaches 59.39 SpotSound mIoU (Table 1), exceeding the prior main-table best by 1.49 points. R1@.3 and R1@.5 improve by 2.75 and 3.00 points, respectively: gains extend from loose overlap to the stricter localization criterion. On Clotho-Moment, NOVA reaches 86.86 mIoU (+1.26), with R1@.3/.5 of 93.49/90.84. On UnAV, NOVA reaches 73.46 mIoU and 89.00/80.00 recall under the released-subset protocol in Section 4.

With direct timestamp generation, RelTwin reaches 59.43 SpotSound mIoU, improving over matched SFT by 1.02 points (CI [0.23, 1.81]). The comparison yields 92 wins, 239 ties, and 69 losses. The replication summary in Table 2(c) gives 59.16 mean mIoU. The following analyses examine how candidate supervision changes the response to inverse queries.

5.2. Timestamp contrasts recover query-specific selection

Paired SFT returns identical answers for 37/160 inverse-query pairs across 22 recordings; RelTwin reduces this to one (Table 2). Of those 37 pairs, 27 become jointly correct: the two predictions recover their respective requested relations. Figure 1 illustrates this query-specific recovery, with each inverse query selecting its corresponding event window.

JointPairAcc increases by 31.25 points (source-group CI [22.35, 40.50]) and Swap error decreases by 32.50. Across all pairs, 51 become jointly correct and one regresses. Relation mIoU increases by 13.97 points (CI [9.84, 17.51]), connecting selection with overlap.

Across three RelTwin seeds, JointPairAcc averages 81.46 and identical-answer counts are 1, 0, and 0, reproducing the

reduction in same-answer collapse.

5.3. Binding gains persist after timing intervention

To test layout-specific silence, we recompose 80 development audios with 1.5 s lead-in, 0.25 s within-window gaps, 3 s between windows, and 2 s tail, preserving waveforms, order, and queries. The frozen matched models evaluate one template (160 queries). Timing changes for 40 BA-first recordings; 40 AB-first recordings remain waveform-identical.

On changed recordings, RelTwin reaches 87.50 JointPairAcc against SFT’s 67.50 (+20.00; CI [2.94, 33.33]); originally they scored 85.00 and 42.50. RelTwin retains a 20-point selection advantage under the altered timing, supporting query–occurrence binding across both silence layouts. Across all 160 queries, SFT and RelTwin score 81.34 and 90.65 mIoU.

5.4. Why locally valid answers matter

Shared excerpts hold event identity fixed while inverse queries exchange the correct window. Detecting both categories supports both candidates; selecting the requested order requires query-conditioned comparison. RelTwin’s reduced same-answer collapse and improved JointPairAcc capture this distinction in the controlled binding test. Timestamp-answer supervision preserves direct generation at inference.

6. CONCLUSION

NOVA learns query–occurrence binding with locally valid timestamp contrasts in RelTwin, complemented by verification and selective refinement. Matched comparisons and timing intervention demonstrate improved query-specific selection. Beyond detecting event presence, grounding benefits from learning which acoustically supported occurrence answers the query.

7. ACKNOWLEDGMENTS

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Compliance with Ethical Standards. We reuse public ESC-50 [14], SpotSound [8], Clotho-Moment and UnAV [2], AudioGrounding [1], and AEGBench [18] data, including programmatically constructed sequences. No new recordings, participant recruitment, human or animal experiments, or listening studies were undertaken.

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